

ACTIVITY 1

Done By:

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Data Wrangling in Python

Code:

```
import pandas as pd
import numpy as np

students = {
    "ID": [101, 102, 103, 104, 105, 105],
    "Name": ["Amit", "Priya", "Rahul", "Sneha", "Riya", "Riya"],
    "Age": [20, 21, 20, 22, 21, 21],
    "Gender": ["M", "F", "M", "F", "F", "F"],
    "Marks": [85, np.nan, 78, 92, np.nan, np.nan]
}

fees = {
    "ID": [101, 102, 103, 104, 105],
    "Pending": [0, 500, 1000, 0, 250]
}

employees1 = {
    "Name": ["John", "Alice"],
    "Age": [25, 30],
    "City": ["Chennai", "Delhi"]
}

employees2 = {
    "Name": ["David", "Sara"],
    "Age": [28, 27],
    "City": ["Mumbai", "Pune"]
}

cars = {
    "Brand": ["Maruti", "Maruti", "Hyundai", "Toyota", "Hyundai", "Toyota"],
    "Year": [2020, 2021, 2020, 2021, 2020, 2021],
    "Sold": [5, 8, 6, 4, 7, 3]
}
```

```
df = pd.DataFrame(students)

print("ORIGINAL DATAFRAME")

print(df)

print("\nMISSING VALUES")

print(df.isnull().sum())

df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

print("\nAFTER FILLING MISSING VALUES")

print(df)
```

AFTER FILLING MISSING VALUES

	ID	Name	Age	Gender	Marks
0	101	Amit	20	M	85.0
1	102	Priya	21	F	85.0
2	103	Rahul	20	M	78.0
3	104	Sneha	22	F	92.0
4	105	Riya	21	F	85.0
5	105	Riya	21	F	85.0

```
df["Gender"] = df["Gender"].replace({"M": 0, "F": 1})

print("\nAFTER REPLACING GENDER")

print(df)
```

AFTER REPLACING GENDER

	ID	Name	Age	Gender	Marks
0	101	Amit	20	0	85.0
1	102	Priya	21	1	85.0
2	103	Rahul	20	0	78.0
3	104	Sneha	22	1	92.0
4	105	Riya	21	1	85.0
5	105	Riya	21	1	85.0

```
filtered = df[df["Marks"] >= 80]

print("\nFILTERED DATA (Marks >= 80)")

print(filtered)
```

FILTERED DATA (Marks >= 80)

	ID	Name	Age	Gender	Marks
0	101	Amit	20	0	85.0
1	102	Priya	21	1	85.0
3	104	Sneha	22	1	92.0
4	105	Riya	21	1	85.0
5	105	Riya	21	1	85.0

```
details = df[["ID", "Name", "Age", "Gender", "Marks"]]
```

```

fees_df = pd.DataFrame(fees)

merged = pd.merge(details, fees_df, on="ID")

print("\nMERGED DATA")

print(merged)

```

```

MERGED DATA
   ID  Name  Age  Gender  Marks  Pending
0  101  Amit   20      0   85.0        0
1  102  Priya  21      1   85.0       500
2  103  Rahul  20      0   78.0      1000
3  104  Sneha  22      1   92.0        0
4  105  Riya  21      1   85.0       250
5  105  Riya  21      1   85.0       250

```

```

car_df = pd.DataFrame(cars)

print("\nCAR DATA")

print("\nGROUP BY YEAR")

print(car_df.groupby("Year").sum(numeric_only=True))

```

```

GROUP BY YEAR
      Sold
Year
2020    18
2021    15

```

```

print("\nGROUP BY BRAND")

print(car_df.groupby("Brand")["Sold"].sum())

```

```

GROUP BY BRAND
Brand
Hyundai    13
Maruti     13
Toyota      7

```

```

print("\nDATA FOR YEAR 2020")

print(car_df.groupby("Year").get_group(2020))

```

```

DATA FOR YEAR 2020
   Brand  Year  Sold
0  Maruti  2020    5
2  Hyundai  2020    6
4  Hyundai  2020    7

```

```

print("\nDUPLICATE ROWS")

```

```
print(df[df.duplicated()])
```

```
no_duplicate = df.drop_duplicates()
```

```
print("\nAFTER REMOVING DUPLICATES")
```

```
print(no_duplicate)
```

	ID	Name	Age	Gender	Marks
0	101	Amit	20	0	85.0
1	102	Priya	21	1	85.0
2	103	Rahul	20	0	78.0
3	104	Sneha	22	1	92.0
4	105	Riya	21	1	85.0

```
emp1 = pd.DataFrame(employees1)
```

```
emp2 = pd.DataFrame(employees2)
```

```
concat_df = pd.concat([emp1, emp2], ignore_index=True)
```

```
print("\nCONCATENATED DATAFRAME")
```

```
print(concat_df)
```

	Name	Age	City
0	John	25	Chennai
1	Alice	30	Delhi
2	David	28	Mumbai
3	Sara	27	Pune